Jeongjin Lee

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RESEARCH INTERESTS

My research interests broadly lie in extreme value theory, with a primary focus on developing novel statistical inference and methodologies for multivariate extremes. My work encompasses copula models, graphical models, geometric extremes, dimension reduction methods, spatial statistics, and environmental statistics.

ACADEMIC EXPERIENCE

Assistant Professor Sep 2026 – Present
Department of Statistics and Data Science
Pukyong National University, Busan, South Korea.

Senior Research Associate July 2024 – Aug 2026
School of Mathematical Sciences, Lancaster University, Lancaster, UK.
Mentor: Jennifer Wadsworth

Postdoctoral Fellow Sep 2022 – June 2024
naXys Institute, Namur & LIDAM/ISBA, UCLouvain, Belgium.
Mentor: Johan Segers and Anna Kiriliouk

EDUCATION

Colorado State University, Fort Collins, CO
Ph.D. in Statistics 2022
Advisor: Daniel Cooley
M.S. in Statistics 2019, earned en route to Ph.D.

Kyungpook National University, Daegu, South Korea
M.S. in Statistics 2016
B.S. in Statistics and Economics - *double major* 2013
Rank: 1/34

AWARDS

Best Oral Presentation Award, Journal of the Korean Data & Information Science Society (2015)
Outstanding Graduate Student Award, Kyungpook National University (2014)
Scholarship for Academic Excellence, Kyungpook National University (2011–2015)
Presidential Award for Statistical Analysis Contest, Kyungpook National University (2013)

WORKING PAPERS

- **Lee, J.**, Corradini, M., and Wadsworth, J. (2026+). Dimension reduction for multivariate geometric extremes.
- **Lee, J.** and Segers, J. (2026+). Geometric properties of Hüsler-Reiss distributions in X-vine models.

UNDER REVIEW

[†]Co-first author (equal contribution, alphabetical order); [*]Corresponding author

- **Lee, J.**[*] and Wadsworth, J. (2025). Geometric criteria for identifying extremal dependence and flexible modeling via additive mixtures. [arXiv] [Code]
- Kim, M.[†] and **Lee, J.**[*†] (2025). Hypothesis testing for partial tail correlation in multivariate extremes. [arXiv] [Code]

PEER-REVIEWED PUBLICATIONS

- **Lee, J.**[*] and Cooley, D. (2026+). Transformed-linear prediction for extremes. *Environmetrics, to appear* [arXiv] [Code]
- Campbell, R.[†], Grolmusova, K.[†], Kakampakou, L.[†], and **Lee, J.**[*†] (2026+). Analysing extreme rainfall via a geometric framework. *Extremes, to appear* [arXiv]
- **Lee, J.** and Kim, Y. (2026). Structure learning for multivariate extremes: A comparative study of regional UK rainfall. *AIMS Mathematics*, 11(3). [DOI]
- Kiriliouk, A.[†], **Lee, J.**[†], and Segers, J.[†] (2025). X-vine models for multivariate extremes. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 87(3), 579–602. [DOI] [Code]
- **Lee, J.**[*], Cooley, D., Wagner, A., and Liston, G. (2025). A calibration method for projecting future extremes via a linear mapping of parameters. *Environmental and Ecological Statistics*, 32, 1–20. [DOI]

APPLIED AND REGIONAL PUBLICATIONS

- **Lee, J.**, Kim, E. S., and Kim, Y. (2025). A prediction model for heatwaves and tropical nights based on apparent temperature. *Journal of the Korean Data & Information Science Society*, 36(6), 1057–1067.
- Hughes, S., Rondeau, M., Shannon, S., Sharp, J., Ivins, G., **Lee, J.**, Taylor, I., and Bendixsen, B. A. (2021). Holistic self-learning approach for young adult depression and anxiety compared to medication-based treatment-as-usual. *Community Mental Health Journal*, 57(2), 392–402.
- **Lee, J.** and Kim, Y. (2016). A spatial analysis of the Neyman-Scott rectangular pulses model using an approximate likelihood function. *Journal of the Korean Data & Information Science Society*, 27(5), 1119–1131.

- Gjonbrataj, J., Choi, W.-I., Bahn, Y. E., Rho, B. H., **Lee, J. J.**, and Lee, C. W. (2015). Incidence of idiopathic pulmonary fibrosis in Korea based on the 2011 ATS/ERS/JRS/ALAT statement. *The International Journal of Tuberculosis and Lung Disease*, 19(6), 742–746.
- **Lee, J.**, Kim, N. H., Kwon, H. J., and Kim, Y. (2014). A Bayesian analysis of return levels for extreme precipitation in Korea. *The Korean Journal of Applied Statistics*, 27(6), 947–958.

PRESENTATIONS

Invited Presentations

- **CMStatistics**, Birkbeck, University of London, UK. Dec 2025
  Geometric criteria for identifying extremal dependence and flexible modeling via additive mixtures
- **Shanghai Center for Mathematical Sciences**, China. July 2024
  X-vine models for multivariate extremes
- **Extreme Value Analysis (EVA) Conference**, Bocconi University, Italy. June 2023
  Partial Tail Correlation for Extremes.
- **Young Researcher Day (YRD)**, UCLouvain, Belgium. Feb 2023
  Partial Tail Correlation for Extremes.
- **Computational and Methodological Statistics (CMStatistics)**, King's College London, UK. Dec 2022
  Partial Tail Correlation for Extremes.
- **University of Namur**, Belgium. Oct 2022
  Partial Tail Correlation for Extremes.

Contributed Presentations & Posters

- **Extreme Value Analysis (EVA) Conference**, UNC Chapel Hill, USA. June 2025
  Identifying Extremal Dependence Classes Using Additive Mixtures in the Geometric Framework
- **ATMS Workshop**, KU Leuven, Belgium. May 2023
  X-Vines: Modelling Extremal Dependence by Vine Constructions for Exponent Measures.
- **Royal Statistical Society of Belgium (RSSB) Annual Meeting**, Belgium. Oct 2022
  Partial Tail Correlation for Extremes.
- **International Centre for Mathematical Meetings (CIRM)**, Marseille, France. Sep 2022
  Partial Tail Correlation for Extremes.
- **Joint Statistical Meetings (JSM)**, Virtual. Aug 2021
  Transformed-Linear Prediction for Extremes.
- **Extreme Value Analysis (EVA) Conference**, University of Edinburgh, Virtual. July 2021

Transformed-Linear Combination of Regularly Varying Random Variables and Linear Prediction for Extremes.

- **Colorado-Wyoming Chapter of the American Statistical Association**, Virtual. Apr 2021
  Transformed-Linear Combination of Regularly Varying Random Variables and Linear Prediction for Extremes.
- **Workshop on Risk Analysis in the Earth System**, Lawrence Berkeley National Lab, USA. (Poster) July 2019
  Extremal Partial Dependence.

TEACHING EXPERIENCE

Pukyong National University

- Introduction to Data Science Fall 2026
- Matrix Algebra Fall 2026

Colorado State University

- **STAT 307:** Introduction to Biostatistics Spring 2020
- **STAT 301:** Introduction to Statistical Methods Fall 2019

Teaching Assistant & Consultant

- **Walk-in Lab Consulting** Spring 2021
- **STAT 204:** Statistics for Business Students (Recitation) Fall 2018, Spring 2019
- **STAT 201:** General Statistics (Recitation) Spring 2018

PEER REVIEW SERVICE

- *Electronic Journal of Statistics*
- *Extremes*
- *Statistics and Computing*

SOFTWARE AND WEB PROFILES

- **Personal Website:** jeongjinlee88.github.io
- **Google Scholar:** Jeongjin Lee
- **Software:** `X-vine` R package, available on GitHub.

TECHNICAL CERTIFICATIONS

- **SAS Certified Advanced Programmer** for SAS 9 2014
- **SAS Certified Base Programmer** for SAS 9 2013

REFERENCES

Johan Segers, Ph.D. jjjsegers@kuleuven.be
Professor
Department of Mathematics
KULeuven

Jennifer Wadsworth, Ph.D. j.wadsworth@lancaster.ac.uk
Professor
School of Mathematical Sciences
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Daniel Cooley, Ph.D. cooleyd@rams.colostate.edu
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Anna Kiriliouk, Ph.D. anna.kiriliouk@uclouvain.be
Assistant Professor
Institute of Statistics, Biostatistics and Actuarial Sciences
UCLouvain